🌿 Cañita Agrotourism Glamping Village – Category I EIA

ESTUDIO DE IMPACTO AMBIENTAL - CATEGORÍA I (EIA LEVEL 1)

Project Title: Proposed 3-Hectare Low-Impact Glamping Village & Agrotourism Hub

Project Location: Corregimiento de Cañita, Distrito de Chepo, Provincia de Panamá, Republic of Panama

Proponent: [Insert Company/Owner Name]

Regulatory Authority: Ministerio de Ambiente (MiAmbiente), Republic of Panama

Legal Framework: Executive Decree No. 123 of 2009 (As amended by Executive Decree No. 40 of 2024 and Executive Decree No. 2 of 2026)

1. Executive Summary

This Environmental Impact Assessment (EIA Category I) addresses the construction and operation of a low-impact eco-tourism and farm-to-table agrotourism village in Cañita, Panama. The project operates entirely off-grid across a 3-hectare (30,000 m²) plot. Infrastructure: 25 prefabricated bamboo/canvas tents; a 4,000 ft² two-story main building (commercial kitchen, reception, offices, board room, employee dormitories, manager quarters); 15–20 kW solar PV + 40–60 kWh $LiFePO_4$ battery; propane cooking; composting toilets (UDDT); 3-stage greywater wetland; spring water extraction limited to 3,000 GPD. Key findings: temporary low-to-moderate construction impacts; zero-discharge wastewater; 2 of 3 hectares dedicated to native reforestation; +3 High Positive socio-economic impact.

1.1 Overview

- **Project Name:** Cañita Eco Glamping & Regenerative Farm
- **Total EIA Study Area:** 3 hectares (nested inside 50 Ha master parcel)
- **Built Footprint:** ~0.4 hectare (250 ft × 150 ft excavated hilltop)
- **Glamping Units:** 25 tents (30 m² each, bamboo + canvas)

- **Main Building:** 4,000 sq ft (2 storey: dining, kitchen, registration, yoga deck)
- **Staff Housing:** On site dormitory (within 3 hectare boundary)
- **Manager Suite:** 1 unit
- **Project Type:** Low impact eco tourism / agro tourism / wellness retreat
- **Development Phase:** Phase 1 of 16 phase / 80 hectare master plan

1.2 Location & Access

- **Region:** Cañita, District of Chepo, Panamá Province
- **Distance from Pan American Highway:** ~2 km north
- **Nearest Landmark:** Centro Educativo Básico General Clímaco Delgado Barahona (500 m)
- **Access Road:** Existing 5 m wide dirt/gravel easement (will be upgraded)
- **Utilities:** Available 400 m away; project will be 100% off grid using solar + propane

1.3 Land Use & Topography (From Appraisal)

- **Total Master Parcel:** 50 hectares (legal Finca 142678)
- **Topography:** 40% undulated / wavy, 60% broken mechanizable terrain
- **Excavated Plateau:** 250 ft × 150 ft (37,500 sq ft) – already cleared hilltop
- **Soil Classification:** Type VI (non arable, suitable for forest, pasture, wildlife)
- **Drainage:** Good natural drainage, clay textured soil
- **Existing Vegetation:** Secondary forest (native trees, shrubs, grasses)

1.4 Sustainability Pillars (10 points)

1. Solar Power: 15 20 kW PV + 40 60 kWh $LiFePO_4$ battery
2. Propane System: Centralized LPG + leak detection
3. Spring Water: 3,000 GPD extraction + 9,000 gal storage + monthly testing
4. Rainwater Harvesting: Catchment → auxiliary irrigation
5. Greywater Recycling: 3 stage (grease trap → wetland → drip irrigation)
6. Composting Toilets: UDDT in all 25 tents + main building
7. Food Waste Composting: On site hot composting
8. Bamboo + Canvas Construction: Low carbon footprint
9. Rattan Furnishings: 100% natural fibers
10. Zero Emission Mobility: Electric ATVs, e bikes, horses

1.5 Agro Tourism & Farm to Table

- **On site production:** vegetables, fruits, melons, fish & shrimp, eggs, goat milk, medicinal herbs
- **Local sourcing:** pork, chicken, beef from Cañita/Chepo ranchers
- **Wellness:** yoga, meditation, hiking, nature study, indigenous healer workshops

2. Master Environmental Impact Matrix (Leopold Method)

This matrix identifies and scales the localized, low-impact variables of the proposed 3-hectare glamping village. It confirms that the project falls squarely within the criteria of a **Categoría I** project under Panamanian law due to its highly localized, mitigable, and temporary negative impacts, paired with long-term positive socio-economic and ecological outcomes.

Component / Activity	Site Clearing	Shallow Drilling	Tent Setup	Solar Install	Farm Ops	Waste Mgmt
Air Quality	-2	-1	-1	0	0	+1
Noise	-2	-2	-1	0	0	0
Soil & Geology	-2	-1	-1	0	+1	+1
Surface/Groundwater	-2	-2	-1	0	-1	+2
Flora & Fauna	-2	-1	-1	-1	+1	0
Socio-Economic	+1	0	+2	+1	+3	+2
Aesthetics	-1	0	-1	-1	+2	0

Impact Scale Key: +3 High Positive | +2 Moderate Positive | +1 Low Positive | 0 No Impact | -1 Low Negative | -2 Moderate Negative | -3 High Negative

3. Solar PV Microgrid & Liquefied Petroleum Gas (LPG) System

To preserve the acoustic ecosystem of the Cañita moist forest, the energy framework eliminates traditional diesel generators under normal operating profiles.

[15-20 kW Ground Solar PV Array] ➔ [MPPT Controllers] ➔ [40-60 kWh LiFePO4 Battery Bank] ➔ [8-10 kW Hybrid Inverters]

Solar PV Array: 15 kW to 20 kW ground-mounted photovoltaic array. Located in optimized clearings to mitigate mature canopy tree removal.

Energy Storage Bank: 40 kWh to 60 kWh Lithium Iron Phosphate ($LiFePO_4$) battery matrix. Selected specifically for tropical climates due to its high thermal runaway threshold and the absolute absence of hazardous heavy toxic metals.

Powerhouse Enclosure (Casa de Máquinas): A separate, secure block-and-mortar structure featuring structural passive ridge ventilation and automated DC extraction fans to ensure internal ambient temperatures remain under 35°C (95°F). Floors are finished with non-conductive, chemical-resistant industrial epoxy.

Thermal Energy Loop (LPG): All high-load commercial heating (kitchen ovens, cooktop burners, and guest hot water systems) is powered exclusively by propane gas via heavy-duty portable tanks.

Propane Gas Yard Architecture: Constructed as an open-air, fenced yard exactly 5 meters away from the primary kitchen. Tanks are anchored to a reinforced concrete pad to prevent seismic displacement and are shielded by a lightweight, non-combustible solar canopy. Distribution pipes utilize seamless Schedule 40 black steel, a low-floor automatic gas leak detector loop, and an exterior emergency automated shut-off solenoid valve.

4. Hydrological Capacity & Water Safety Plan (3,000 GPD)

4.1 Water Balance Budget & Storage Allocations

The property extracts and manages a maximum volume of 3,000 Gallons Per Day (GPD) split between an evaluated natural spring and a rainwater harvesting retention lake network.

Potable Allocation (1,500 GPD): Calculated at 60 GPD per tent for 25 units (assuming double occupancy at 30 gallons per person per day). Supplies guest showering, hand-washing, and treated drinking infrastructure.

Non-Potable Allocation (1,500 GPD): Supplies the commercial agrotourism kitchen, restaurant cleaning cycles, and primary organic agricultural irrigation.

Storage Infrastructure: A matrix of three 3,000-gallon UV-resistant, BPA-free polyethylene tanks (9,000 gallons total capacity, providing 3 full days of operational autonomy). Tanks are gravity-optimized on elevated terrain adjacent to the powerhouse.

4.2 Spring Extraction & Wellhead Protection Guidelines

Core Extraction Drilling: Utilizes portable, light-footprint drilling rigs over the spring eye. The process enforces a total ban on synthetic polymers or chemical drilling muds, utilizing 100% clean water for mechanical cooling. The upper 2 meters of the borehole casing are sealed with high-density bentonite clay grout to prevent vertical surface water contamination.

Hydrological Safeguards (Protection Zones):
Immediate Exclusion Zone (0–15 Meters): Surrounded by an absolute physical fence perimeter around the concrete collection box (Caja de Captación). Zero structures, human traffic, or waste management lines are permitted.

Controlled Buffer Zone (15–50 Meters): Limited exclusively to distribution plumbing and footpaths. Glamping accommodations and wastewater networks are strictly prohibited.

Ecological Monitoring: A non-resettable turbine mechanical flow meter is integrated into the collection box line. Daily data logs ensure the 3,000 GPD extraction cap is maintained, keeping extraction under 20% of the spring’s dry-season baseflow to protect the native ecosystem.

4.3 Kitchen & Commercial Greywater Treatment Train

[Kitchen & Main Structure Greywater] ➔ [3-Chamber Grease Trap] ➔ [Sub-Surface Constructed Wetland] ➔ [Drip Irrigation]

Mechanical Grease Trap: Commercial kitchen lines from the main structure discharge into a subsurface three-chamber trap to separate food solids and isolate floating fats, oils, and greases (FOG) before entering the core filtration network.

Phytoremediation Basin: Effluent passes into a basin (0.6m deep) lined with an impermeable 30-mil EPDM membrane. The bed is backfilled with graded volcanic river gravel and planted with alternating rows of Canna Lilies (Canna indica) and King Tut Papyrus (Cyperus papyrus).

Hydraulic Rule: The waterline sits permanently 10cm below the gravel surface. Zero open standing water is present, eliminating mosquito vectors and odors.

Storm Surge Overflow Bypass: Features a 4-inch perforated PVC skimming weir set 5cm above the gravel line. During torrential downpours (temporales), excess clean stormwater enters the skimmer and routes via a 6-inch gravity pipe to a rock-armored infiltration ditch in Zone 1.

5. Bamboo Urine-Diverting Dry Toilets (UDDT)

Double-Vault Basements: Beneath each of the 25 guest outhouses and the dedicated restroom facilities in the main two-story structure are water-tight, side-by-side concrete block vaults coated inside with an impermeable elastomeric membrane. Zero soil contact.

Urine Diversion: Liquid waste is diverted forward into an isolated gravity line routing to a closed evapotranspiration bed. Solid waste drops directly down into the active concrete chamber.

Vector Controls: Features a 4-inch black PVC thermal solar ventilation chimney extending 50cm above the roofline. All openings and structural bamboo slats are lined with marine-grade stainless-steel insect mesh.

Thermophilic Operations (O&M): Users drop one cup of dry carbon material (bamboo sawdust/rice husks) down the solids vault after each use. Housekeeping staff log core pile temperatures weekly. The pile must maintain a minimum temperature of 55°C (131°F) for 15 consecutive days to ensure total pathogen death. Full vaults undergo a solar-heated 6-to-9-month curing cycle, converting into clean humus used exclusively to fertilize Zone 3 Perimeter Bamboo.

6. Environmental Baseline (Línea Base)

6.1 Physical Environment

Climate: Tropical Moist Forest (Bosque Húmedo Tropical) life zone. Characterized by high humidity, a distinct dry season (January–April), and an intense rainy season (May–December) with frequent storm cells (temporales).

Topography & Soils: Gently sloping terrain draining towards local tributaries. Soils are primarily clay-loams with a high vulnerability to compaction and erosion if heavy machinery is deployed.

Hydrology: An active, clean groundwater table surfacing as a natural freshwater spring on the property, which serves as the primary hydraulic anchor for the development.

6.2 Biological Environment

Flora: Second-growth tropical forest intermixed with degraded pastureland from historic cattle grazing. Primary native tree indicators include regenerating Espavé and Cedro Espino.

Fauna: High avian and insect biodiversity. Sighted species include local parakeets, hummingbirds, small reptiles, and transient mammal populations moving along the Chepo/Bayano biological corridors.

6.3 Socio-Economic Environment

Demographics: The project area impacts the rural community of Cañita, Distrito de Chepo. The local economy is traditionally dependent on subsistence agriculture, logging, and small-scale livestock management. Public services are limited, making off-grid self-sufficiency a necessity.

7. Mitigation Plan (PAMA)

Construction: Geotextile silt fences, manual clearing, daytime noise restrictions (8am-4pm).

Operation: Zero-discharge wastewater loop, weekly digital temperature logging of composting vaults, weekly grease trap cleaning. 100% local hiring policy.

8. Reforestation Zoning

2 hectares preserved: Zone 1 (0.25 ha) – strict native reforestation around spring (Espavé, Sotacaballo, Higuerón).

Zone 2 (0.5 ha) – ornamental corridors (Heliconias, Almendro de Montaña).

Zone 3 (0.75 ha) – Guadua bamboo windbreak + Cedro Espino, Caoba.

Zone 4 (0.5 ha) – food forest (Cacao, Guanábana, citrus, root vegetables). 80% survival guarantee.

9. Environmental Education & Public Awareness

Free bi-monthly STEM field trips for local schools (MEDUCA curriculum). Weekend workshops for visitors/university students (permaculture, alternative sanitation, off-grid microgrids).

10. Emergency Plan (PRE)

LPG leak: Automated floor sensors activate exterior solenoid shut-off, 50m evacuation, dry chemical/CO_2 extinguisher. Call Chepo Bomberos line 103.

Battery thermal event: Ceiling-mounted Stat-X/FirePro aerosol canisters activate at 70°C, cut DC fans, rapid shutdown switch – no water.

Monthly water quality (MINSA): Turbidity

11. 18-Month Project Timeline

Months 1-3: Permitting, EIA submission, water concession, municipal approvals.

Months 4-6: Dry season – pin-piles, pre-cast piers, spring drilling, concrete vaults, 9,000-gal tank matrix.

Months 7-12: Rainy season – prefab tent assembly, main structure framing, planting zones, powerhouse, wetland excavation.

Months 13-15: Propane yard, solar+battery wiring, staff training (thermophilic logs, hospitality).

Months 16-18: Soft launch, MINSA lab tests, first booking.

11. Preliminary Budget ($754,500 USD)

Category	Description	Cost (USD)
1. Pre-Construction & Permitting	MiAmbiente filing, water study, Chepo permits, surveying	$32,000
2. Site Preparation & Utilities	Spring capping, 9,000-gal tanks, 20 kW solar + 60 kWh $LiFePO_4$, powerhouse, grease trap, wetland liner, 26 UDDT vaults, propane yard, trail clearing	$156,000
3. Structural Construction	25 prefab cabins ($11k each), pre-cast piers, 4,000 sq ft main building, kitchen appliances, board room IT	$452,000
4. Landscaping & Reforestation	Zone 1&3 native trees/bamboo, Zone 2&4 food forest, planting labor	$34,500
5. Contingency Reserves	Wet-season delay, MINSA lab escrow, local payroll padding	$80,000
TOTAL		**$754,500**

13. Project Description (Detailed)

Project Name: Cañita Eco-Glamping & Regenerative Farm

Total EIA Study Area: 3 hectares (nested inside 50-Ha master parcel)

Built Footprint: ~0.4 hectare (250 ft × 150 ft excavated hilltop)

Glamping Units: 25 tents (30 m² each, bamboo + canvas)

Main Building: 4,000 sq ft (2 storey: dining, kitchen, registration, yoga deck)

Staff Housing: On-site dormitory (within 3-hectare boundary)

Manager Suite: 1 unit

Project Type: Low-impact eco-tourism / agro-tourism / wellness retreat

Development Phase: Phase 1 of 16-phase / 80-hectare master plan

Location: Cañita, Chepo, Panamá – 2 km north of Pan-American Highway.

Topography: 40% undulated, 60% broken terrain. Type VI soil. Good drainage, secondary forest.

Sustainability Pillars (10): Solar, propane, spring water, rainwater, greywater recycling, UDDT, food composting, bamboo/canvas, rattan, zero-emission mobility.

Agro-Tourism: Vegetables, fruits, fish/shrimp, eggs, goat milk, medicinal herbs + local rancher partnerships. Wellness: yoga, meditation, hiking, indigenous workshops.

14. Environmental Management Plan (PMA) Matrix – Structured Table

Component	Impact	Mitigation	Indicator	Responsible	Frequency
Soil & Geology	Erosion	Geotextile fences, manual footings, gravel, terracing	No ruts, sediment logs	Env. Consultant	Weekly
Water Resources	Spring contamination	3-stage greywater, UDDT, 15m exclusion, 30m buffer	Lab tests pass	Project Operator	Monthly
Flora & Fauna	Habitat fragmentation	Live fences, no cutting large trees, boardwalks, downward LED	Tree count, wildlife logs	Project Operator	Quarterly
Air & Noise	Dust, noise	Water spray, daylight hours, electric tools, no diesel	Noise logs	Site Manager	Weekly/monthly
Socio-Economic	Community friction	70% local hiring, staff housing, school programs	Payroll records	Project Admin	Monthly
Solid Waste	Pollution	Pack In Pack Out, composting, recycling, zero single-use plastic	Compost vs landfill kg	Site Manager	Daily/weekly
Energy & GHGs	Emissions	100% off-grid solar + $LiFePO_4$, propane cooking, no diesel	Solar logs, zero diesel	Energy Tech	Monthly

Component	Impact	Mitigation	Indicator	Responsible	Frequency
Aesthetics	Visual intrusion	Low-profile bamboo/canvas, solar on hilltop, native vegetation	Photo records	Design Manager	Annually

15. Technical Infrastructure Tables

Solar + Battery: 15-20 kW ground PV, $LiFePO_4$ 40-60 kWh, 8-10 kW inverter, aerosol fire suppression.

Water Supply: 3,000 GPD from spring, 9,000 gal storage, 15m exclusion zone, monthly E. coli testing.

Greywater: 3-chamber grease trap → subsurface wetland (Canna/Papyrus) → drip irrigation.

Composting Toilets: UDDT, dual vault, ≥55°C for 15 days, 6-9 month curing.

Rainwater: Catchment from canvas roofs → 3,000 gal storage → auxiliary irrigation.

Propane: Centralized LPG yard ≥5m from kitchen, Sch 40 pipe, leak detectors, automatic shut-off.

Solid Waste: On-site hot composting + recycling station + Pack In Pack Out.

16. Socio-Economic & Community Plan

Local hiring: 70%+ from Cañita/Chepo; on-site staff housing (zero commuting).

Indigenous collaboration: Paid healers and guides with written agreements.

Education: Free bi-monthly STEM field trips for local schools; guest workshops.

Local supply chain: Direct purchase from local farmers, ranchers, fishermen.

Micro-enterprise: Bayano Lake excursions (70/30 split), artisan sales (100% to artisans).

Grievance mechanism: Complaint log + quarterly community meetings.

Adventure Excursions: Shuttle services, guided hiking, monkey expeditions, river rafting, Bayano caves, bird watching, kayaking, fishing charters, San Blas affiliate program.

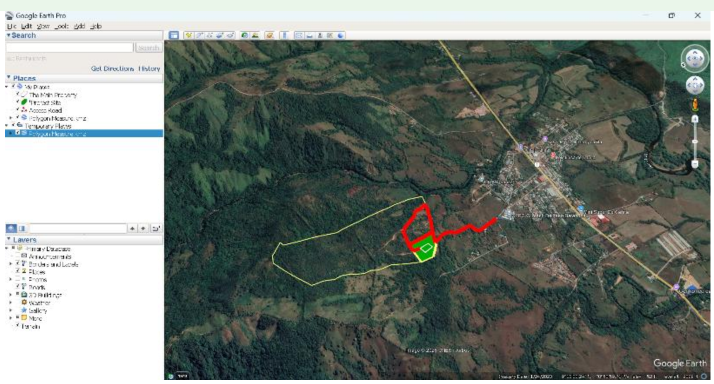
17. Legal Checklist & Approval Sequence

Documents client must obtain: Memorial Petitorio (notarized), Sworn Declaration, promoter’s ID copy, Public Registry certificate, property deed (Folio Real), cadastral map, Paz y Salvo, evaluation fee receipt, Land Use Certification (MIVIOT), water concession permit (if required).

Approval pipeline: ANATI title verification → MIVIOT zoning → Hire MiAmbiente consultant → On-site verification → EIA submission to MiAmbiente (Chepo) → 30-60 day review → Environmental Resolution → Municipal construction permit.

Critical permits: MiAmbiente registered consultant, local lawyer, fire department approval for propane, felling permit (if any tree >10 cm DBH).

18. Images



Yellow = 50 ha boundary, Red = access road,
Green = 3-ha zone, White = 250×150 ft plateau.

Daytime view – hilltop layout, solar panels, central lodge.

Nighttime view – warm LED lamps on pathways and decks.